



BIG DATA

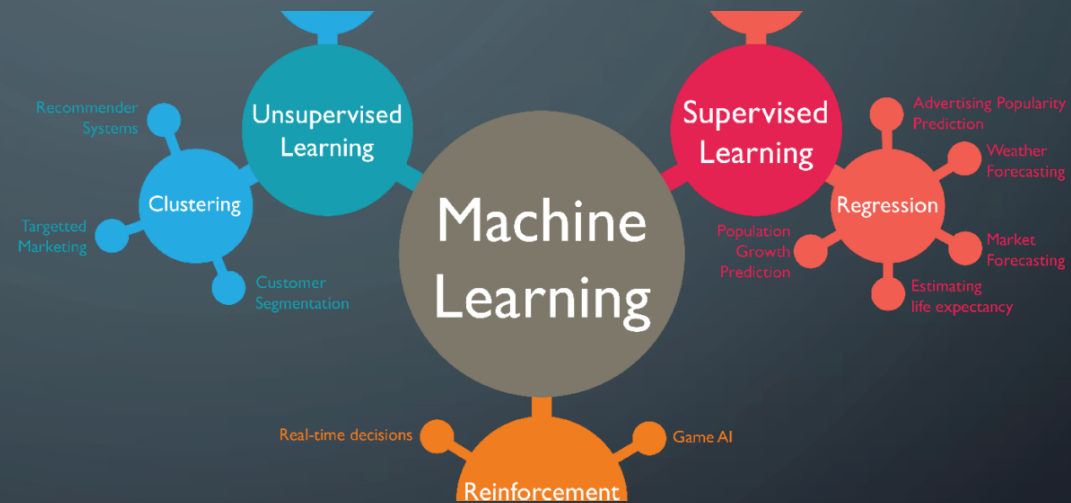
MACHINE LEARNING

LEARNING OBJECTIVES

- At the end of this presentation, you should be able to
 - distinguish supervised from unsupervised learning algorithms;
 - explain the principles of supervised learning;
 - understand how to use training and testing sets;
 - understand how to assess classifier quality using accuracy, sensitivity and specificity;
 - diagnose overfitting.

WHAT IS MACHINE LEARNING?

- Supervised Learning
 - Classification
 - Prediction (Regression)
- Unsupervised Learning
 - Clustering
- Reinforcement Learning
 - Games
 - Robotics



SUPERVISED VS UNSUPERVISED

- Unsupervised learning

- Our data doesn't have the answer.

length	ears	snout
50cm	pointy	5cm
25cm	pointy	2cm
30cm	floppy	3cm

- Supervised learning

- Our data has the answer.

answer

type	length	ears	snout
dog	50cm	pointy	5cm
cat	25cm	pointy	2cm
dog	30cm	floppy	3cm

training data

aka 'tagged' data

VARIABLES

dependent

independent variables



type	length	ears	snout
dog	50cm	pointy	5cm
cat	25cm	pointy	2cm
dog	30cm	floppy	3cm

SUPERVISED LEARNING

- Why do we learn when we already know the answer?
 - Supervised learning is more successful than unsupervised
 - People 'annotate' a small subset of the data to be able to use supervised learning
 - The annotated data helps learn the model, which can be applied to new data.

- Supervised learning
 - Our data has the answer.

type	length	ears	snout
dog	50cm	pointy	5cm
cat	25cm	pointy	2cm
dog	30cm	floppy	3cm

↑
answer

aka 'tagged' data

SUPERVISED LEARNING - STEPS

- Task: Use software to find information in data.
1. Feed the software data that holds the information.
 2. "Train a model"
 - finds relevant patterns in the data
 3. Use the model to find the answer in new data.

attribute
or feature



type	length	ears	snout
dog	50cm	pointy	5cm
cat	25cm	pointy	2cm
dog	30cm	floppy	3cm

TRAINING AND TESTING SETS

.... for supervised learning

- Training set: 70 – 80% of data
- Testing set: 20 – 30% of data

type	length	ears	snout
dog	50cm	pointy	5cm
cat	25cm	pointy	2cm
dog	30cm	floppy	3cm
cat	25cm	pointy	1cm
cat	38cm	pointy	2.5 cm
dog	30cm	floppy	4cm
cat	32cm	floppy	1.5cm



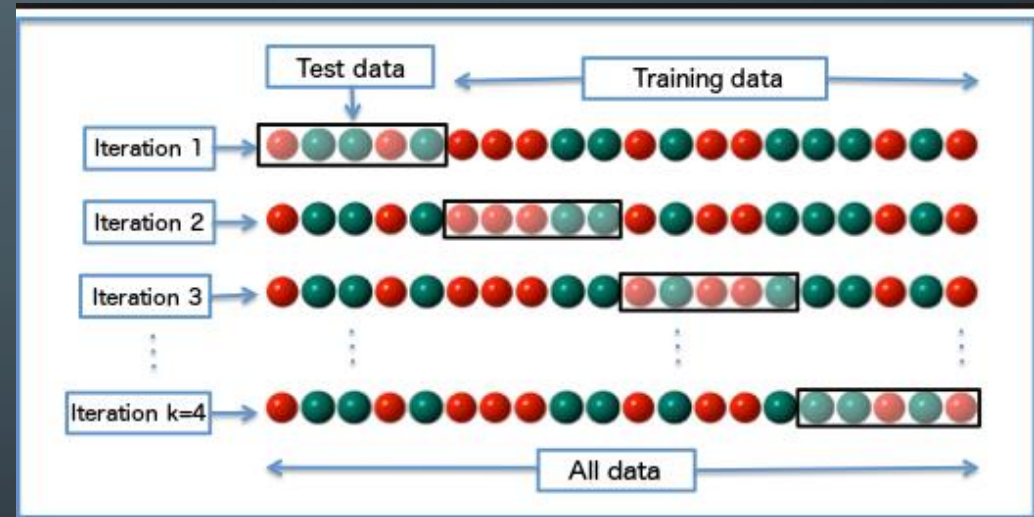
type	length	ears	snout
dog	45cm	pointy	5cm
dog	35cm	floppy	5.5cm
cat	32cm	floppy	2cm

30 samples =
'statistically significant'

CROSSVALIDATION

.... for supervised learning

- Subdivide data into k 'folds'
 - k is often set to 10
 - Stratified folds = balanced
 - Each iteration, use a different test set. All other data is used for training.
- Retain the model that leads to the best classification of the test set.



Source: Wikipedia

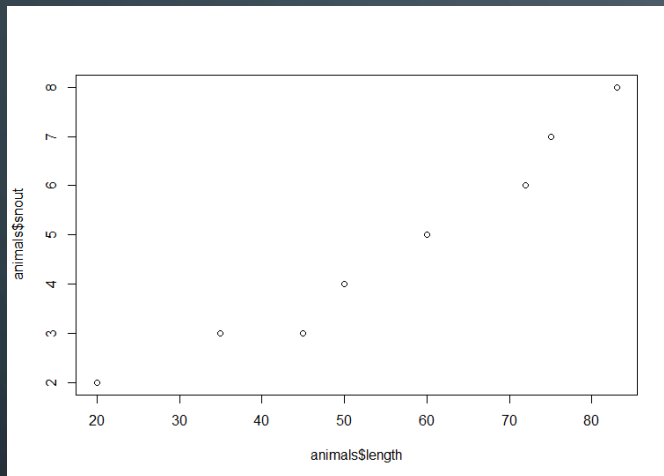
UNSUPERVISED LEARNING

- **Clustering** can be used to group items
 - We have to specify the number of clusters we want.
 - We rely on the variance in the data.
 - If we have very many variables, we may have to find the ones that 'covary' (are correlated).

length	ears	snout
50cm	pointy	5cm
25cm	pointy	2cm
60cm	floppy	6cm

CORRELATION

positive correlation



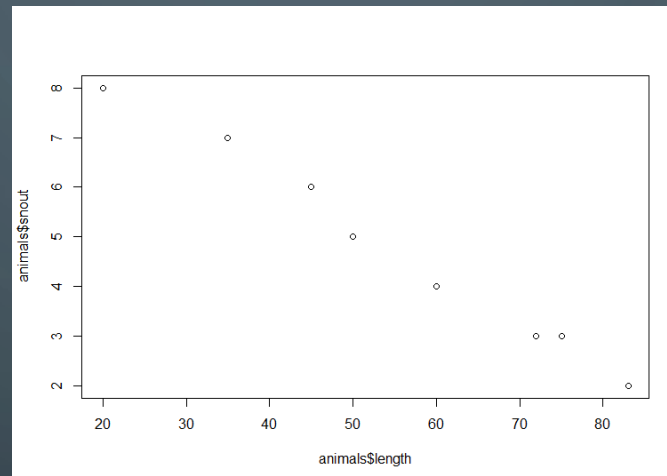
Correlation coefficient

weak correlation: <0.3

moderate correlation: $0.3 - 0.7$

strong correlation: >0.7

negative correlation



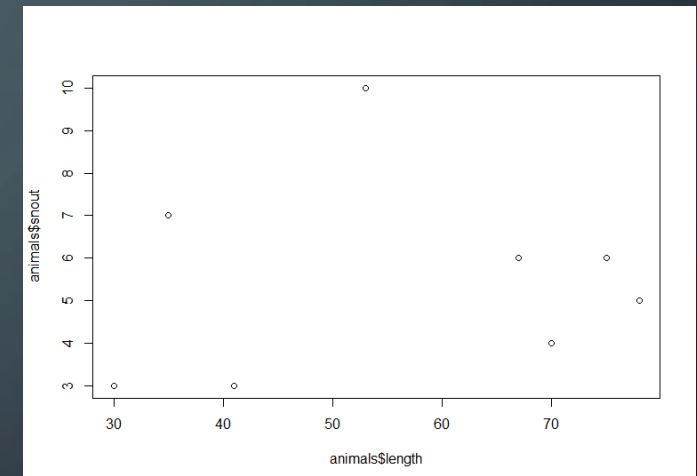
Correlation coefficient

weak correlation: >-0.3

moderate correlation: $-0.3 - -0.7$

strong correlation: <-0.7

no correlation



CLASSIFICATION RESULT: CONFUSION MATRIX

Prediction (from ML)	Reference (actual)		
		dog	cat
	dog	TP	FP
	cat	FN	TN

Prediction (from ML)	Reference (actual)		
		dog	cat
	dog	70	2
	cat	3	40

TP = true positive
FP = false positive
FN = false negative
TN = true negative

MEASURES OF QUALITY: ACCURACY

Prediction	Reference		
		dog	cat
	dog	TP	FP
	cat	FN	TN

TP = true positive

FP = false positive

FN = false negative

TN = true negative

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN}$$

ACCURACY, BEST

	Reference		
		dog	cat
	Prediction		
	dog	73	0
	cat	0	42

$$\text{Accuracy} = \frac{73+42}{73+0+0+42}$$
$$= 1$$

ACCURACY, WORST

	Reference		
		dog	cat
	Prediction		
	dog	0	66
	cat	54	0

$$\text{Accuracy} = \frac{0+0}{0+66+54+0}$$
$$=0$$

MEASURES OF QUALITY: SENSITIVITY

	Reference		
		dog	cat
	dog	TP	FP
	cat	FN	TN

TP = true positive
FP = false positive
FN = false negative
TN = true negative

$$\text{Sensitivity} = \frac{TP}{TP+FN}$$

MEASURES OF QUALITY: SPECIFICITY

	Reference		
		dog	cat
	dog	TP	FP
	cat	FN	TN

TP = true positive
FP = false positive
FN = false negative
TN = true negative

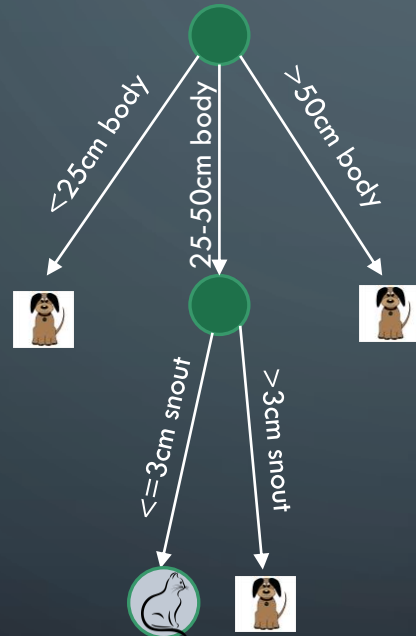
$$\text{Specificity} = \frac{TN}{FP + TN}$$

OVERFITTING

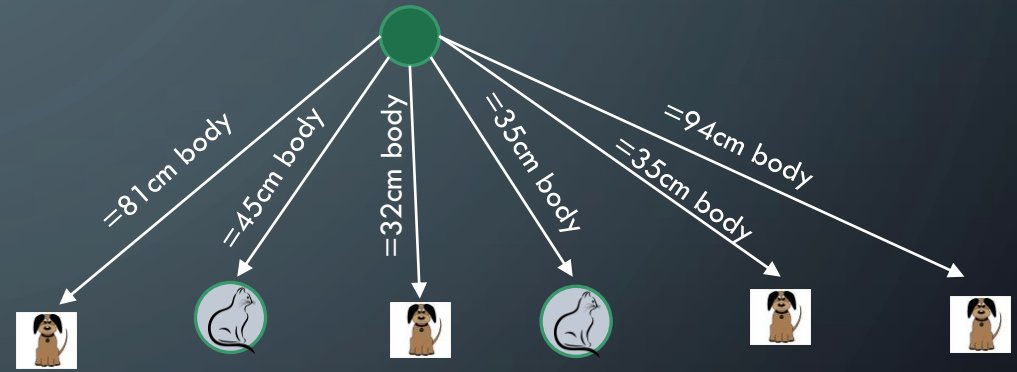
Training

type	length
dog	81cm
cat	25cm
dog	94cm
cat	45cm
cat	38cm
dog	32cm
cat	35cm

Good model:



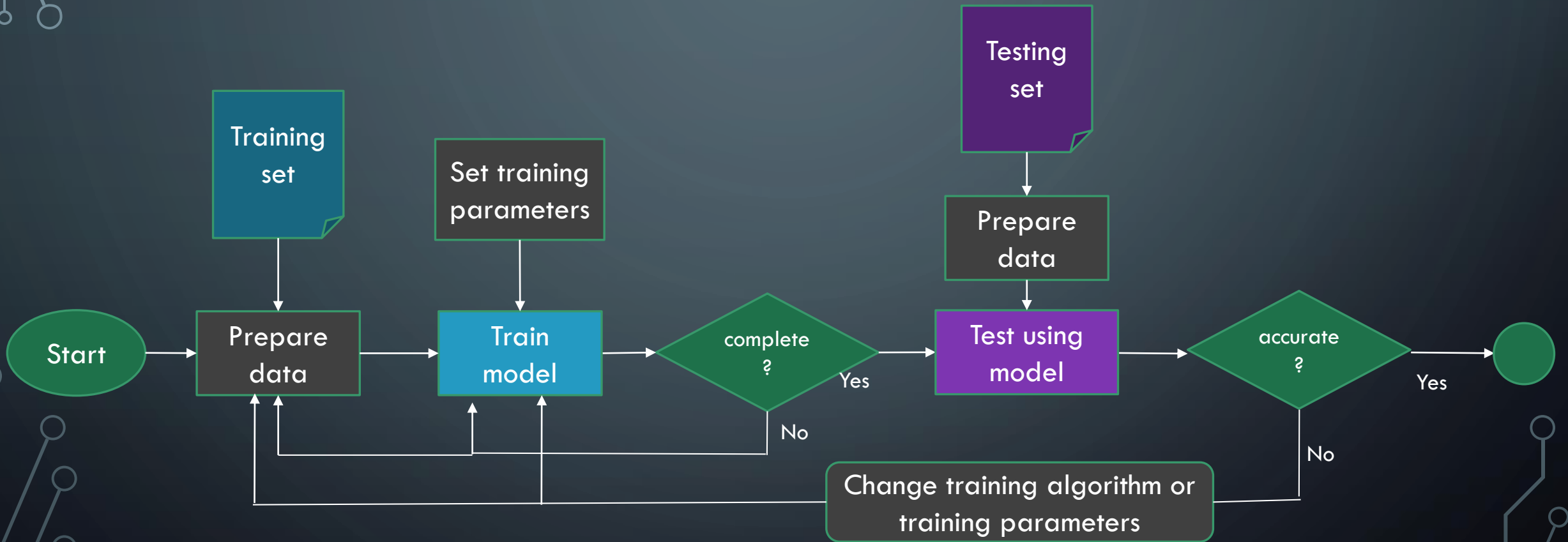
Bad model:



Testing

type	length
dog	45cm
dog	35cm
cat	32cm

SUPERVISED LEARNING PROCESS



SUMMARY

- Machine learning has supervised and unsupervised methods (as well as reinforcement learning).
- Supervised learning is more accurate than unsupervised learning but needs data with the correct answer supplied.
- Supervised learning uses a large part of the data to train a model. It then tests the model on data the model has not seen before.
- The quality of the model can be measured in terms of accuracy, sensitivity and specificity.